



NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY
K H U L N A

A System Design for AI Prompt Engineering Learning Hub

Course Code: CSE4104

Course Title: Software Development III

Team: CSE4104-7C-T02

Department of Computer Science and Engineering

Northern University of Business and Technology, Khulna

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Team Information Sheet

Team Name:	CSE4104-7C-T02
Section:	7C
Project Title:	AI Prompt Engineering Learning Hub
GitHub Repository Link:	https://github.com/RahminHossain/cse4104-7c-t02-ai-prompt-engineering-learning-hub
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Team Members Roles & Credentials

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1. Project Information

Project Title: AI Prompt Engineering Learning Hub

1.1 Project Overview

The AI Prompt Engineering Learning Hub is a web-based educational platform designed to help users learn and practice prompt engineering for generative AI tools. It provides an interactive environment where users can write prompts, test them, and receive real-time feedback from an AI system.

1.2 Problem Statement

Many users struggle to write effective prompts for AI systems. Current learning resources like blogs and videos are static and do not provide hands-on practice or feedback. As a result:

1. Users rely on trial and error.
2. Outputs are often low-quality.
3. There is no centralized platform for structured learning.

This project aims to solve these issues by creating an interactive and intelligent learning system.

1.3 Objectives

1. Develop an interactive platform for learning prompt engineering
2. Provide automated AI-based prompt evaluation and feedback
3. Create sandbox environments for testing prompts
4. Introduce gamification (XP, badges, challenges) to improve engagement

2. Architecture Diagram

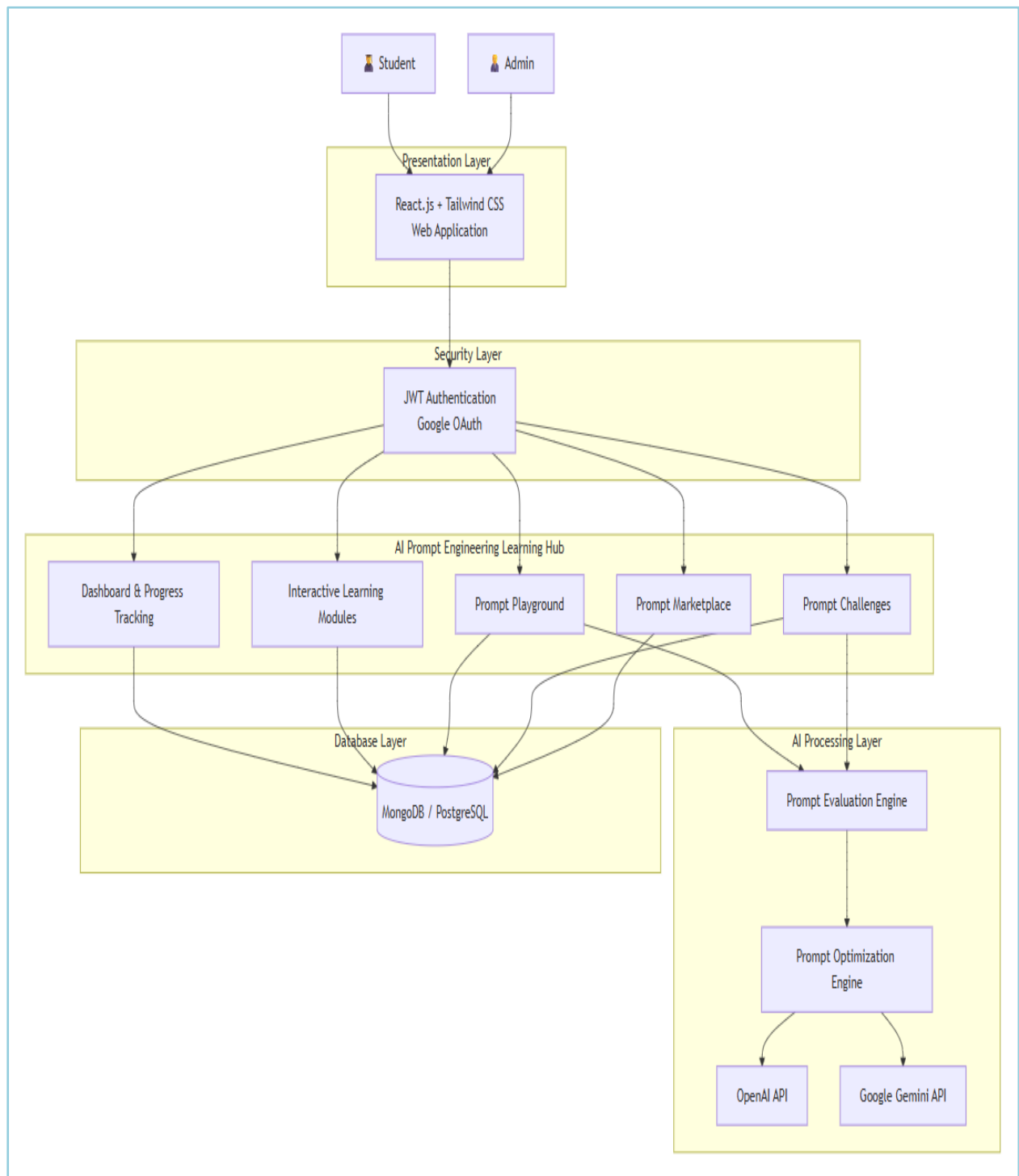


Figure 1: AI Prompt Engineering Learning Hub — System Architecture

3. ER Diagram

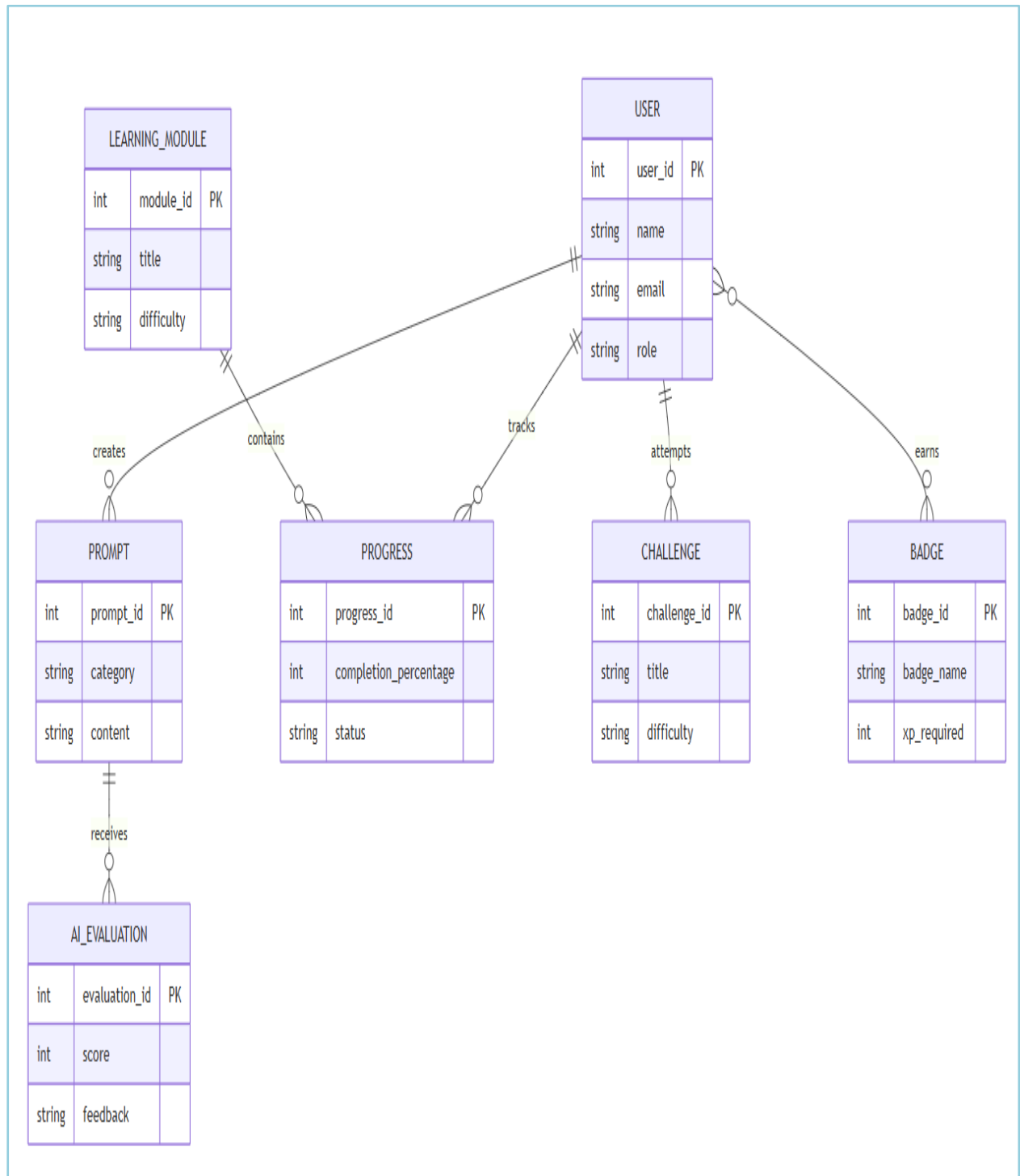


Figure 2: AI Prompt Engineering Learning Hub — ER Diagram

4. Use Case Diagram

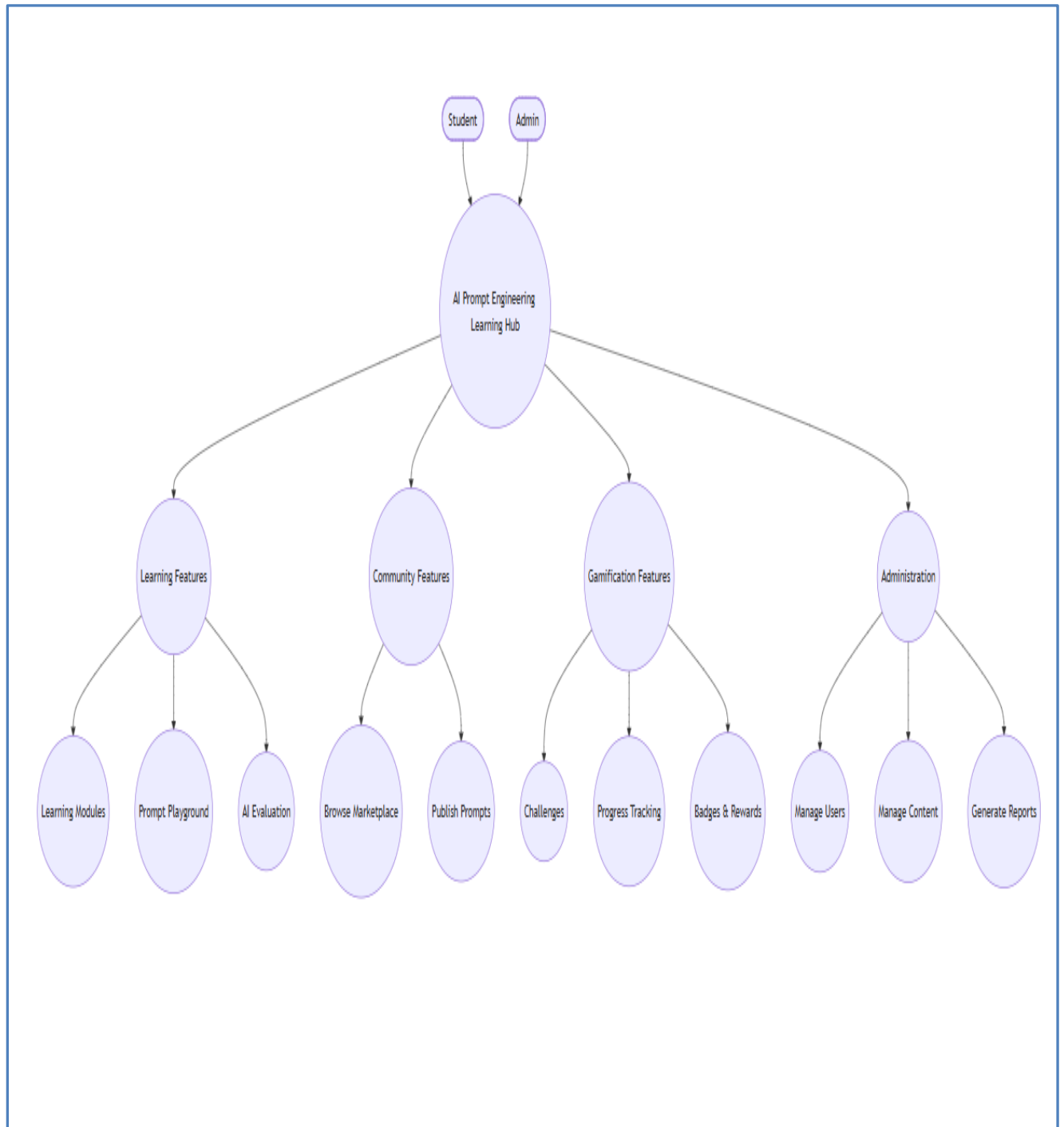


Figure 3:AI Prompt Engineering Learning Hub— Use Case Diagram

5. Activity Diagram

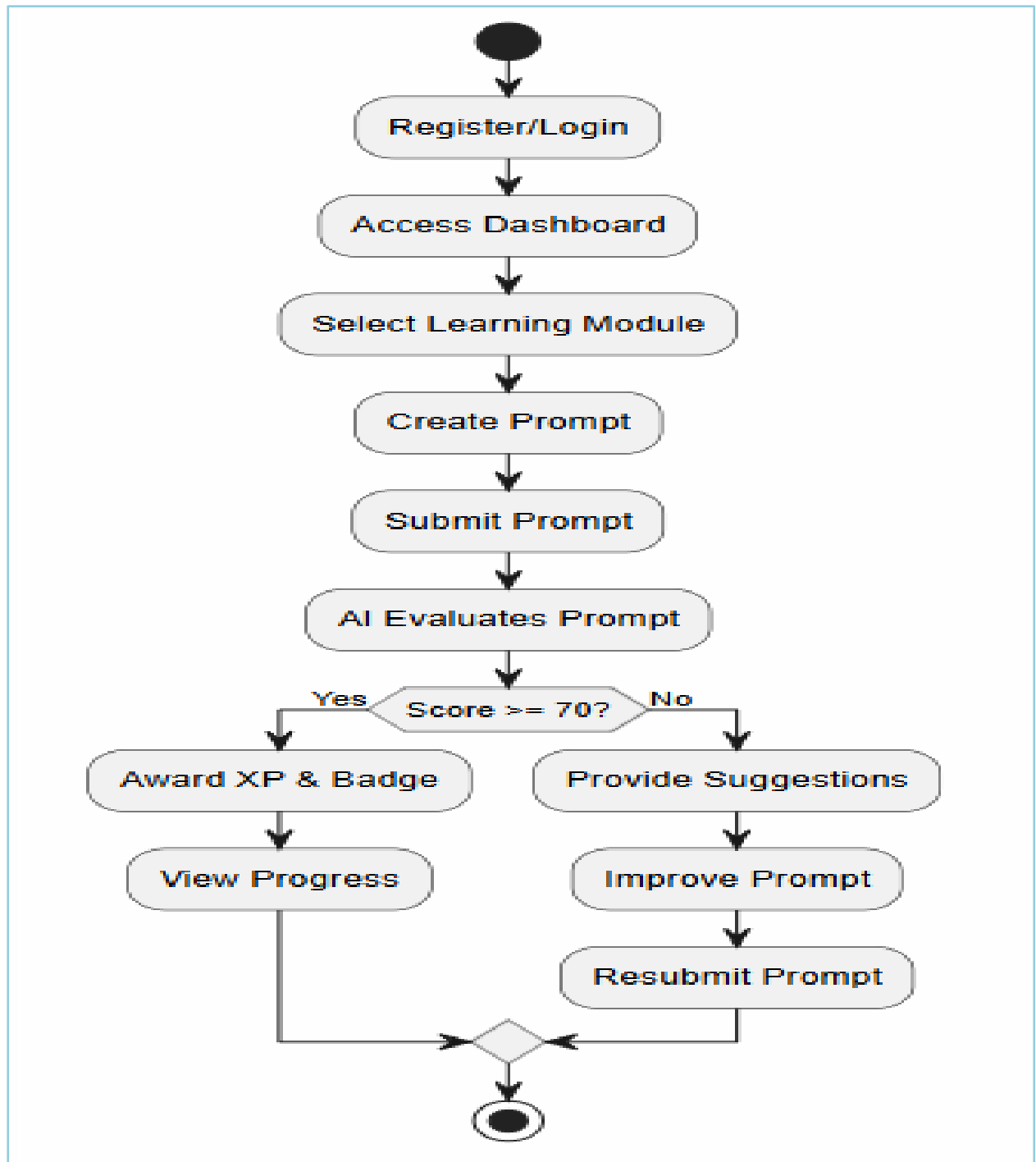


Figure 4: AI Prompt Engineering Learning Hub— Activity Diagram

6. API Design

The AI Prompt Engineering Learning Hub will use a RESTful API architecture to facilitate communication between the frontend application, backend services, database, and AI integration layer. The following APIs are planned based on the functional requirements identified in the Software Requirements Specification (SRS).

6.1 Authentication APIs

API Endpoint	Purpose	Input Data	Expected Output
POST /register	Register a new user account	Name, Email, Password	Registration success message and User ID
POST /login	Authenticate users using email and password	Email, Password	Authentication token and user information
POST /google-login	Authenticate users using Google OAuth	Google authentication credentials	Authentication token and user profile
POST /forgot-password	Allow users to reset forgotten passwords	Email address	Password reset confirmation
POST /logout	End user session securely	Authentication token	Logout confirmation

6.2 User Dashboard APIs

API Endpoint	Purpose	Input Data	Expected Output
GET /progress	Retrieve user learning progress	User ID	Progress percentage and completed modules
POST /progress/update	Update user progress after completing activities	User ID, Activity Information	Updated progress status
GET /badges	Retrieve earned badges and XP points	User ID	Badge list and XP statistics
GET /bookmarks	Retrieve bookmarked prompts	User ID	Saved prompt collection

6.3 Learning Module APIs

API Endpoint	Purpose	Input Data	Expected Output
GET /modules	Retrieve all learning modules	None	List of learning modules
GET /modules/{id}	Retrieve details of a specific module	Module ID	Module content and details
POST /quiz/submit	Submit quiz or assessment answers	Quiz answers	Quiz score and feedback

6.4 AI Prompt Playground APIs

API Endpoint	Purpose	Input Data	Expected Output
POST /prompt/evaluate	Submit prompt for AI evaluation	Prompt text	Prompt score and evaluation feedback
POST /prompt/optimize	Generate an optimized version of a prompt	Prompt text	Improved prompt version
GET /prompt/history	Retrieve previously evaluated prompts	User ID	Prompt evaluation history

6.5 Reverse Prompt Engineering Challenge APIs

API Endpoint	Purpose	Input Data	Expected Output
GET /challenges	Retrieve available challenges	None	Challenge list
POST /challenge/submit	Submit challenge solution	Challenge ID, User Response	Challenge score and feedback

6.6 Social Marketplace and Export APIs

API Endpoint	Purpose	Input Data	Expected Output
GET /marketplace	Retrieve public prompt marketplace entries	Optional search keyword	List of prompts
POST /marketplace/publish	Publish prompt to marketplace	Prompt content, Tags, Category	Publication confirmation
POST /marketplace/upvote	Upvote a marketplace prompt	Prompt ID	Updated vote count
GET /prompt/export	Export prompt for external use	Prompt ID	Exportable prompt format

7. AI Integration Workflow

7.1 Why AI is Needed

Traditional educational systems and technical instruction frameworks rely heavily on static rule-engines, string matching, or basic regular expression (regex) parsing scripts. These deterministic methodologies are completely incapable of measuring structural abstract configurations, semantic natural language alignment, target safety thresholds, or qualitative intent behind complex human-generated sentences.

Because the primary mission of the AI Prompt Engineering Learning Hub is to dynamically evaluate and guide users through production-grade engineering concepts—such as Few-Shot Learning frameworks, Role-play allocation mechanics, and multi-layered Chain-of-Thought execution trees—the platform infrastructure must utilize AI natively. An embedded Large Language Model (LLM) abstraction framework acts as an automated, real-time "Prompt Critic". It programmatically assesses structural composition, computes mathematical quality scores (0-100), maps complex failure modes, and synthesizes optimized programmatic alternatives instantly—tasks a traditional backend code loop cannot perform.

7.2 Which AI Service/Model Will Be Used

To preserve production-grade reliability, secure data routing, and optimal token parsing speeds, the integration schema utilizes enterprise cloud software development kits (SDKs) to form a dual-provider orchestration engine:

- **Primary Engine Core (Google Gemini API):** The platform adopts Gemini 1.5 Pro for multi-layered context processing and deep conceptual grading, alongside Gemini 1.5 Flash to manage low-latency structural scoring routines inside the real-time interactive playground.
- **Cross-Evaluation & Failover Layer (OpenAI API SDK):** The system uses GPT-4o to validate evaluation score matrices, mitigate generative model biases during competitive engineering challenges, and ensure architectural redundancy.

7.3 Input and Output to the AI System

Data pipes traversing the backend orchestration layer are managed via strict JSON boundaries across two primary functional environments:

A. Real-Time Playground & Critic Engine

- Input to the AI System:
 - User Payload: Raw text input composed by the student inside the browser sandbox environment.
 - System Instructions (Contextual Anchors): A protected, background contextual wrapper instructing the target LLM to act as an objective technical grading rubric, forcing structural validation rules and enforcing a raw JSON output block.
- Output from the AI System:
 - The processing backend captures and interprets a structured programmatic output object layout inside a shaded container box:

```
{
  "composite_score":84,
  "error_breakdown":{
    "missing_constraints": "The prompt lacks clear boundary limitations.",
    "role_assignment": "Weak structural identity defined for the persona."
  },
}
```

```
"optimized_variation": "Act as a Senior QA Automation Engineer. Review the following code segment under strict memory-efficiency constraints..."
}
```

B. Reverse Prompt Engineering Challenge Lab

- Input to the AI System:
 - Target Objective State: Pre-compiled structured output data (e.g., a specific JSON matrix or structural text layout) generated by the system lesson.
 - User Submission: The prompt text constructed by the user attempting to make the model reproduce that exact output state.
- Output from the AI System:
 - A distance-vector calculation matrix representing raw semantic similarity.
 - A Boolean completion indicator (true / false) that unlocks experience points (XP) and pushes badge rewards directly to the student dashboard.

7.4 Architectural Pipeline

The lifecycle of a single interactive verification run flows sequentially through the application tiers as detailed below:

Step 01: User Question

The student enters a prompt string into the React.js frontend interface and clicks the "Evaluate" button.

Step 02: Backend API

The Express.js backend receives the request, processes the client data securely, attaches necessary context, and forwards the parameters to the AI service.

Step 03: OpenAI / Gemini API

The cloud-based AI models analyze the prompt, evaluate its linguistic structure and constraints, and generate a performance score.

Step 04: Generated Response

The system returns a structured JSON response containing evaluation results, identified errors, and grading metrics.

Step 05: Frontend Display

The frontend application processes the returned data and displays the results in real-time on the user's dashboard interface.

7.5 How AI Improves the Project

- **Actionable Skill Transformation:** Shifts student behavior away from casual, conversational natural language patterns toward precise, programmatic, and deterministic framework-driven interactions.
- **Continuous Feedback Loops:** Removes human-dependency for assignment grading, establishing automated round-the-clock training scenarios with context-aware analysis.
- **Transparent Performance Metrics:** Provides explicit clarity when a text design fails, generating mathematical attributes representing precise alignment flaws, system constraint breaches, or token leakage.
- **Rigorous Core Scope:** Elevating artificial intelligence into a core evaluation, grading, and optimization subsystem ensures that the architecture easily satisfies the non-trivial, complex design standards demanded by the CSE4104 software engineering curriculum.

8. References

1. Prompt Engineering & Generative AI

Artificial Intelligence: A Modern Approach

Stuart Russell & Peter Norvig

→ Covers foundational AI concepts, useful for understanding intelligent systems.

Language Models are Few-Shot Learners

Brown et al., 2020

→ Explains how large language models respond to prompts.

Chain-of-Thought Prompting Elicits Reasoning

Wei et al., 2022

→ Important for advanced prompt engineering techniques.

2. Large Language Models & APIs

OpenAI Documentation

→ <https://platform.openai.com/docs>

→ Used for GPT-based prompt evaluation and feedback.

Google DeepMind (Gemini API)

→ <https://ai.google.dev>

→ Used for Gemini 1.5 Pro & Flash integration.

3. Software Engineering & System Design

Software Engineering

→ Covers SRS, system architecture, and design models.

Designing Data-Intensive Applications

→ Useful for backend architecture and API design.

4. Web Development Technologies

React Documentation

→ <https://react.dev>

→ Used for building the frontend UI.

Express.js Documentation

→ <https://expressjs.com>

→ Used for backend API development.

MongoDB Documentation

→ <https://www.mongodb.com/docs>

→ Used for storing user data, prompts, and progress.

5. REST API Design

RESTful Web APIs

→ Helps design scalable API endpoints.

6. Educational Technology & Gamification

Gamification

Deterding et al., 2011

→ Supports use of XP, badges, and challenges.

The Impact of Gamification on Learning

→ Shows how gamification improves engagement.

7. Additional Online Resources

OpenAI Prompt Engineering Guide

→ <https://platform.openai.com/docs/guides/prompt-engineering>

Google AI Prompt Design Guide

→ https://ai.google.dev/docs/prompt_best_practices